

Global Renewable Energy in 2025: How Far Have We Come?

Global renewable energy trends across primary energy and electricity

75% of Global GHG Emissions Still Come from Fossil Fuels

~75%

of global greenhouse gas emissions come from fossil fuels used for energy

5M+

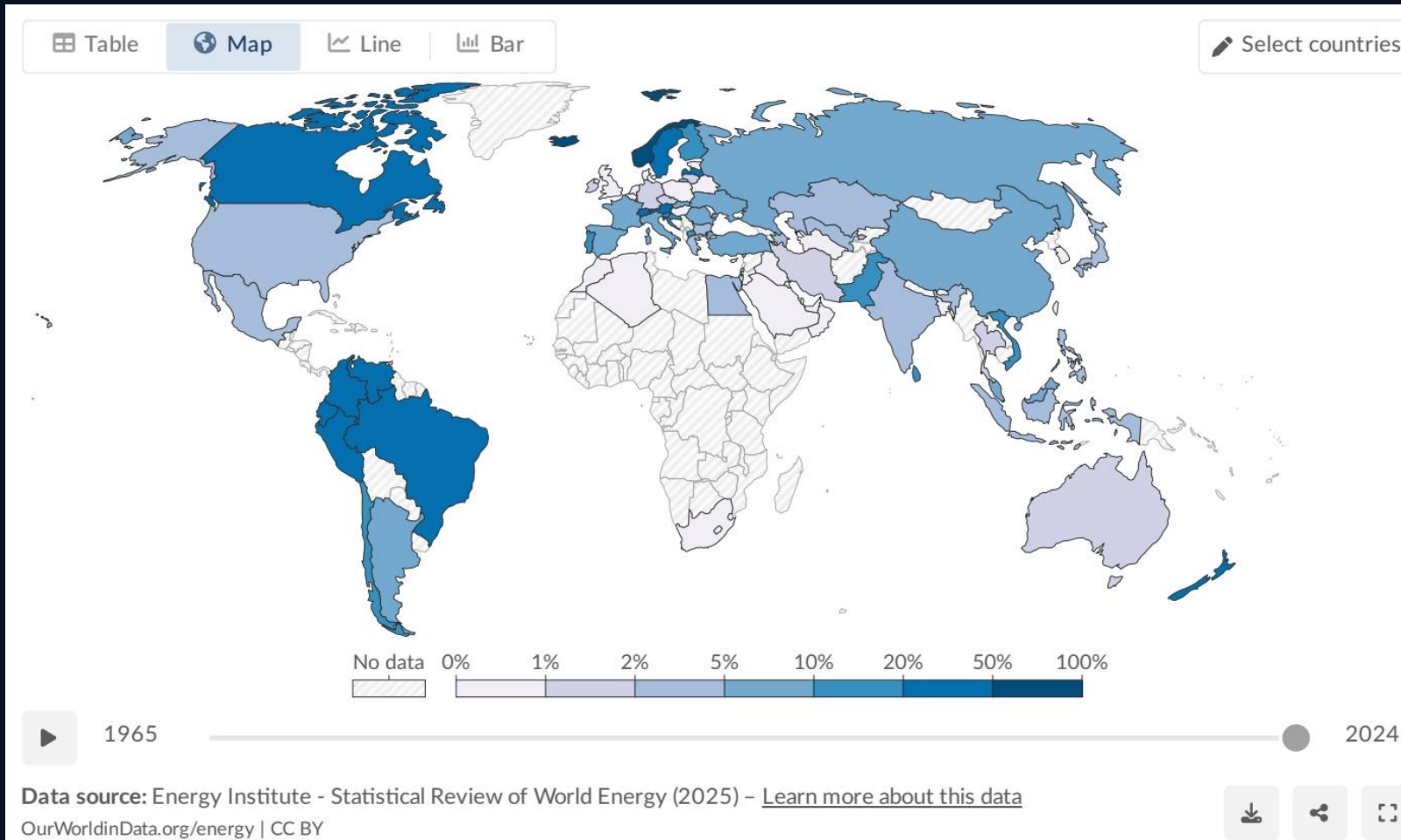
premature deaths per year linked to fossil fuel air pollution

Decarbonizing energy is central to cutting CO₂ and air pollution.

Low-carbon supply options include renewables and nuclear.

Modern renewable metrics typically exclude traditional biomass.

Renewables Now Supply ~14% of Global Primary Energy



~1 in 7 units of global primary energy is renewable (2024).

Primary energy includes electricity, transport, and heating.

Transport & heating remain harder to decarbonize, keeping the share lower than electricity.

Share of primary energy from renewables by country (2024, substitution method)

Source: Our World in Data – Renewable Energy

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar



2024 renewable generation (approx.): hydro ~4,500 TWh; wind ~2,400; solar ~2,000; other ~700.

Wind + solar combined now nearly match hydropower output.

Wind and solar accelerated sharply since ~2010.

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.

Our World
in Data

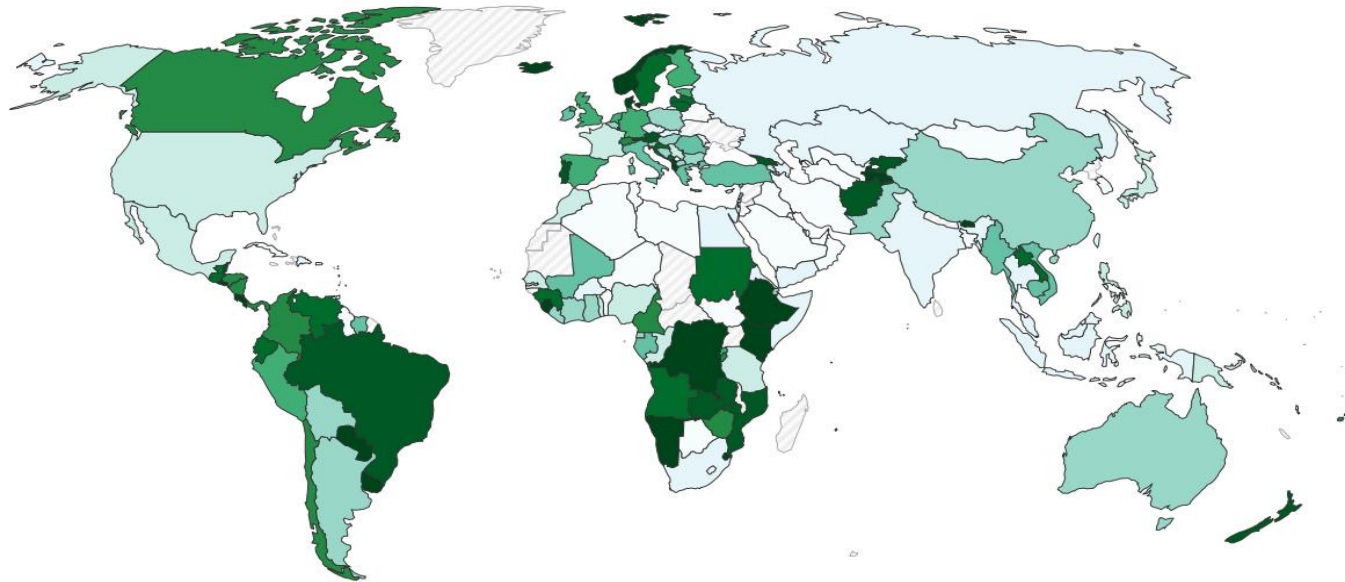
Table

Map

Line

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Select countries



No data 0% 10% 20% 30% 40% 50% 60% 70% 80% 90% 100%

1985

2025

Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)

OurWorldinData.org/energy | CC BY



Share of electricity from renewables by country (2025)

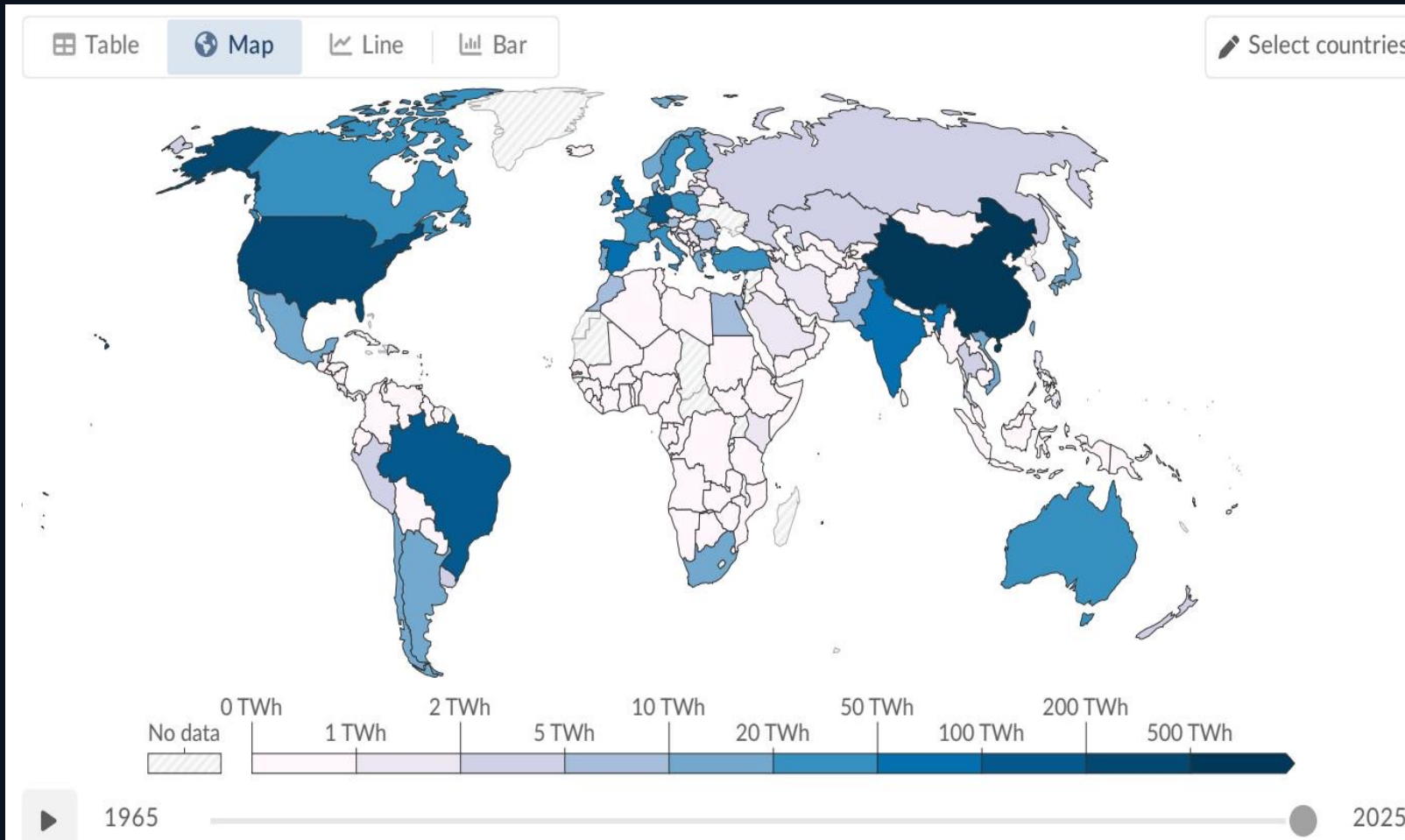
Source: Our World in Data – Renewable Energy

Selected	Renewable share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

Global average is ~30% (2025).

Leaders cluster in hydro-rich regions; laggards include many fossil-fuel producers.

Hydropower Remains the Largest Renewable Source at ~50% of Generation



Hydropower contributes ~47% of renewable electricity (~4,500 TWh in 2024).

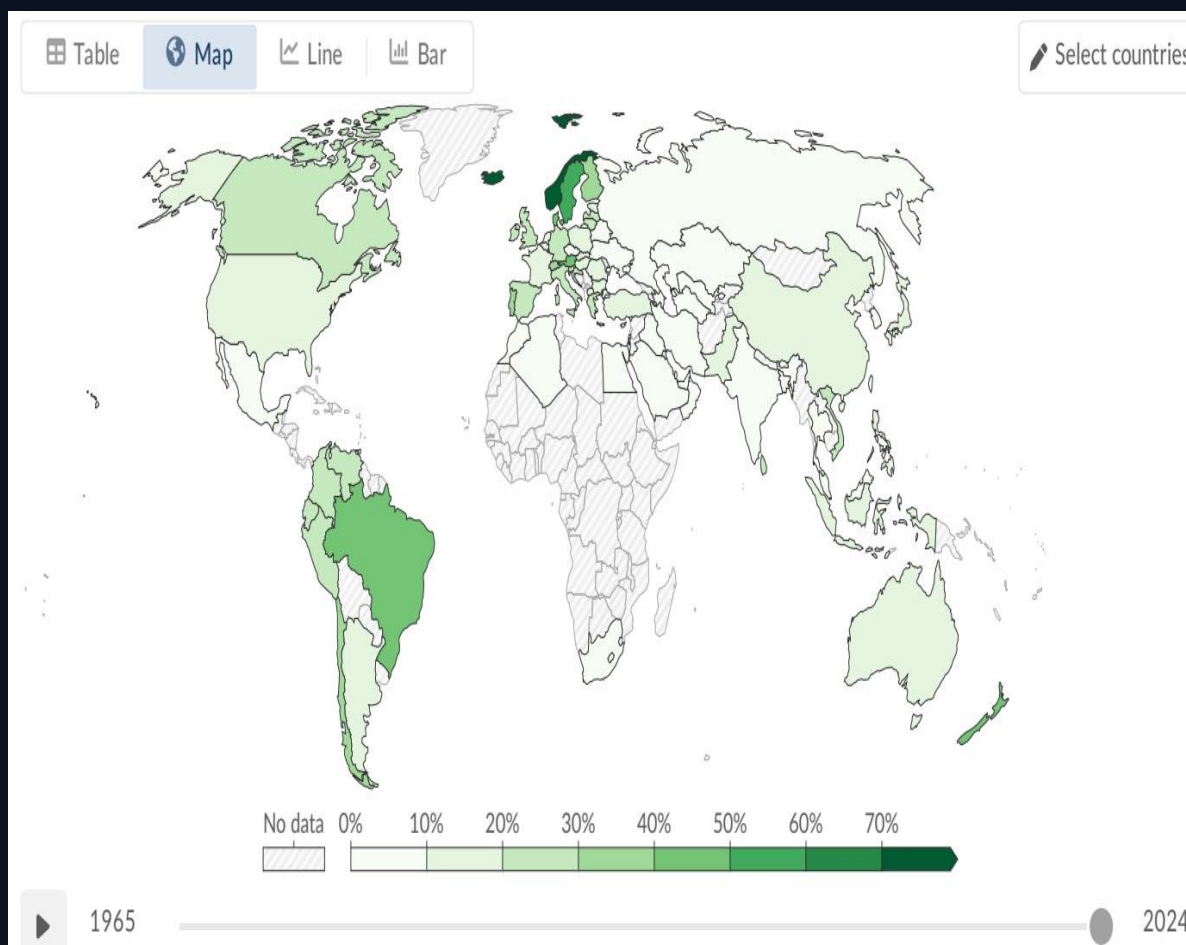
Top producers include China, Brazil, and Canada.

Long-established technology; growth potential often constrained by geography and permitting.

Hydropower generation by country (2025)

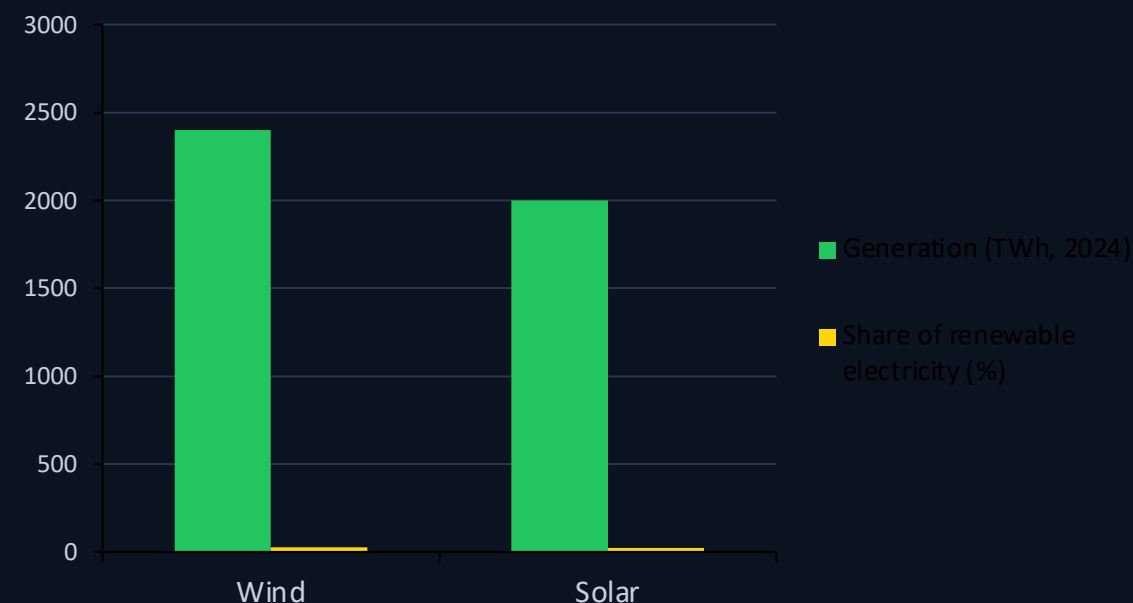
Source: Our World in Data – Renewable Energy

Wind and Solar Are the Fastest-Growing Energy Technologies in History



Wind power generation by country (2025)

Source: Our World in Data – Renewable Energy

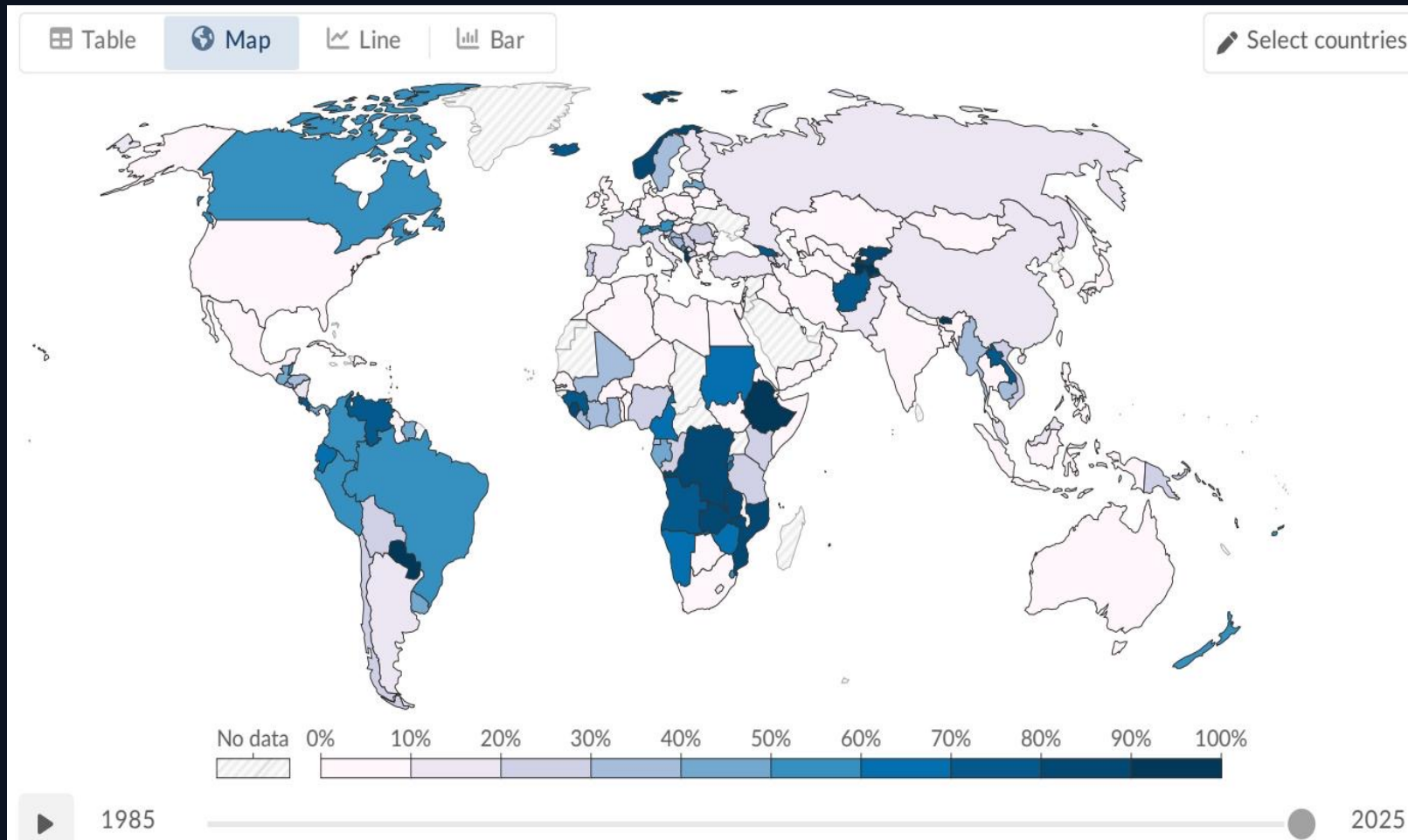


Wind: ~2,400 TWh (~25% of renewable electricity).

Solar: ~2,000 TWh (~21%); identified as the fastest-growing source since 2010.

Together, wind + solar now nearly match hydropower output.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead



Hydro-rich countries (e.g., Scandinavia, Brazil, Canada) reach 65–98% renewable electricity.

Many African and Middle Eastern countries remain below 10% renewables.

Policy, financing, and grid buildout are decisive for catching up.

Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

~14%

of global primary energy is renewable (2024)

~30%

of global electricity is renewable (2025)

Hydropower still provides ~half of renewable electricity, but wind and solar are closing the gap.

Wind (~2,400 TWh) and solar (~2,000 TWh) are scaling quickly; solar is the fastest-growing source.

Regional disparities remain large, with many countries still below 10% renewable electricity.

Transport and heating lag behind electricity—keeping primary energy progress slower.

Priority actions: expand grids, accelerate permitting, mobilize capital, and sustain policy support.